

→ Storing information which is needed again & again

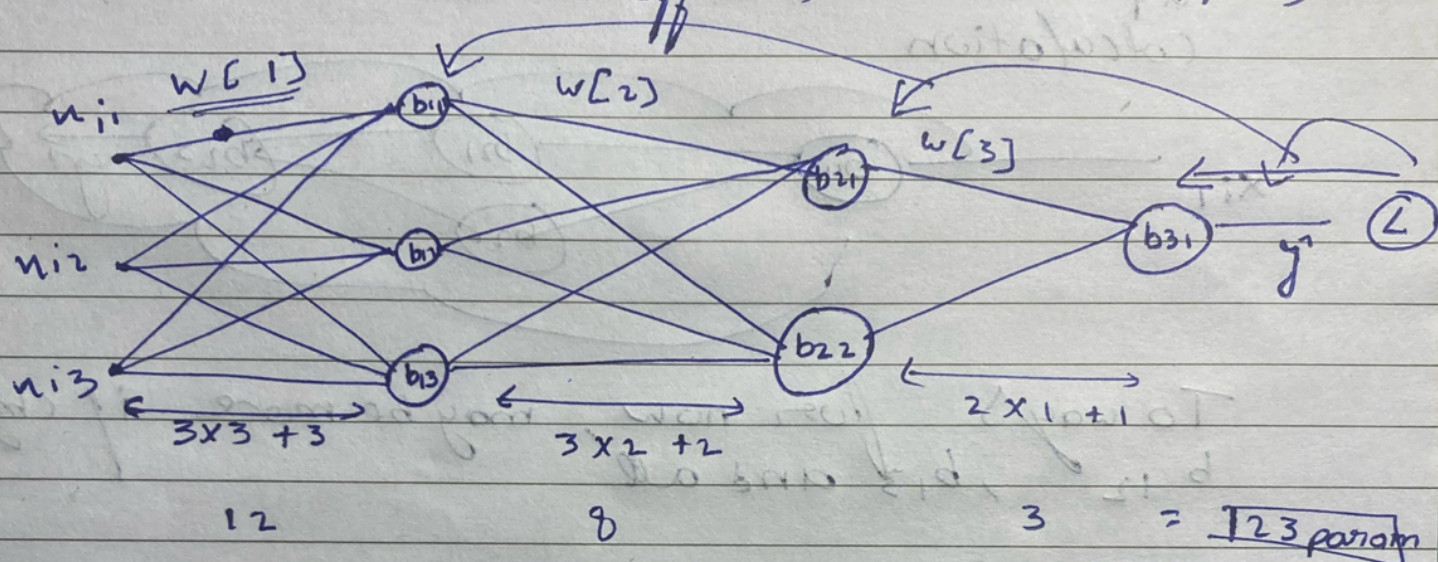
MLP Memorization

Back propagation

Chain rule + Memorization

↓ diff (maths)

↓ CSE (6 pole)



If need to find weight w_{11}^3 then

$$\frac{\partial L}{\partial w_{11}^3} = \frac{\partial L}{\partial y} \times \frac{dy}{dw_{11}^3}$$

→ Because when we are

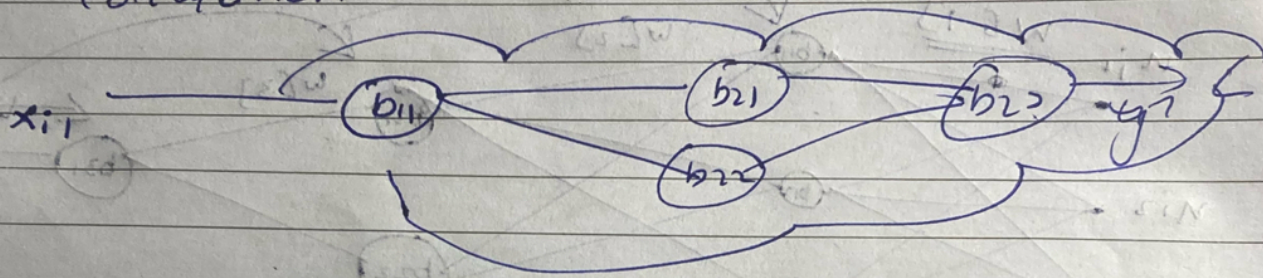
finding loss & update weights with respect to loss then we back propagate & find partial derivatives

$$\frac{\partial L}{\partial w_{11}^2} = \frac{\partial L}{\partial y} \times \frac{dy}{dw_{11}^2} \times \frac{dw_{11}^2}{dw_{11}^3}$$

→ we are taking only one factor for now

updating
for finding the weights of first layer
we need to back propagate

If we are considering all the chances of
update then it will be more complex
calculation



To ways for now, may be more if change
 b_{12} , b_{13} and all

$$x \begin{cases} \rightarrow f(n) \\ \rightarrow g(n) \end{cases} \rightarrow h(f(n), g(n)) = \frac{dh}{dn} = \frac{\partial h}{\partial f(n)} \times \frac{df(n)}{dx} + \frac{\partial h}{\partial g(n)} \times \frac{dg(n)}{dx}$$

$$\frac{dh}{dn} = \frac{\partial h}{\partial f(n)} \times \frac{df(n)}{dx} + \frac{\partial h}{\partial g(n)} \times \frac{dg(n)}{dx}$$

Now

$$\frac{dL}{dw_{11}} = \frac{dL}{dy_1} \left[\frac{dy_1}{do_{21}} \times \frac{do_{21}}{do_{11}} \times \frac{do_{11}}{dw_{11}} + \frac{dy_1}{do_{22}} \times \frac{do_{22}}{do_{11}} \times \frac{do_{11}}{dw_{11}} \right]$$

See we need this much heavy calculation $\frac{dL}{dw_{11}}$
So we are use memorization to store
every partial derivation for future use to reduce
time of calculation